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Health Policy Analysis

Do Medicine Shortages Reduce Access and Increase Pharmaceutical Expenditure? A Retrospective Analysis of Switzerland 2015-2020

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ABSTRACT

Objectives: We analyze how shortages led to changes in access to and expenditure for pharmaceutical care in the Swiss health system between 2015 and 2020.

Methods: We combined cross-sectional and longitudinal data to study medicine shortages by incidence, duration, intensity, and pharmaceutical expenditure. We assessed 4119 markets defined by active ingredient, dosage form, and strength. We classified markets by essential medicine status and other characteristics. We differentiated shortages by the degree to which alternative options are still available. We investigated the first lockdown period of the pandemic, considering also the shortage of COVID-19–specific medicines.

Results: A total of 1964 markets never reported shortages, and 1336 markets reported some shortages; 819 markets reported shortages lasting at least 14 days. Markets with a higher number of manufacturers, a lower co-payment share, and lower prices more frequently reported shortages. We did not find differences by essential medicine status. In 50% of instances, the average price of substitutes available was lower than the price of the product on shortage. The total pharmaceutical expenditure attributed to shortages increased by CHF 17.00 million (€15.63 million) in 2018.

Conclusions: Medicine shortages have substantially reduced access to pharmaceuticals. Switzerland has experienced shortages on a scale similar to that in other countries. Prices of substitutes available at the time of shortages can be higher or lower, indicating an unelastic demand for medicines.

Keywords: essential medicines, market characteristics, supply chain.

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Introduction

A shortage of medicines refers to a situation in which the supply becomes so limited or the demand increases such that excess demand exists for a certain period.[1] The causes of shortages are rooted in demand- and supply-side factors and often are complex in terms of how they affect healthcare.[2] Many shortages are caused by natural disasters, issues in manufacturing quality, and supply-chain interruptions.[3,4] Once a health system experiences a shortage, healthcare providers must adjust medication or treat patients otherwise. Over the past 5 to 10 years, the number of medicines reported to be short has increased sharply, partly because countries have initiated shortage reporting obligations.[5-7] In 2019, the Federal Office for National Economic Supply (Bundesamt für wirtschaftliche Landesversorgung [BWL]) reported 238 events of interruptions in medicine supply in Switzerland.[8] The COVID-19 pandemic has led to short-term spikes in demand for medicines, and trade interruptions have increased public awareness of the need to maintain an overview of the availability of essential medicines.[9]

Medicine shortages are prevalent in many therapeutic areas and not confined to essential medicines as defined by the World Health Organization.[1,10] Shortages force physicians to adjust treatment, substitute medicines, or interrupt pharmaceutical care. This can lead to adverse events for patients and affect the cost of treatment.[11] Often, shortages concern issues with providing sufficient quality that relates to the supply chain and production of medicine.[3,4] Shortages frequently occur in mature multisource markets in which the brand of a once patented medicine and one or several equivalent generic versions are available.[12]

The literature on access to pharmaceutical care and the consequences for expenditure has focused on factors that determine initial access after market entry of a product but not during shortage periods. This is typically pursued by analyzing launch delay, the adoption of new active ingredients, or the availability of generic medicines.[13] Shortages affect people's access by the evenness of availability of medicines.[14,15] Some options, either brand name or a generic version, may still be available. Variations in the evenness of access may be characterized by the incidence, duration, and intensity of shortages. Another dimension of access is ability to pay because patients may need to switch to more expensive or less expensive products.

Studies that use repository data, addressing predominantly North America and to some degree Europe, identified the scope of shortages.[5-7,16,17] Factors associated with how medicine shortages reduce access include the duration of shortages, the therapeutic and product-specific areas affected, and product prices. For the United States, even after the introduction of the Safety and Innovation Act in 2012, the duration of shortages was not reduced.[16] Acute care medicines are subject to relatively longer shortage periods than nonacute care medicines. In ambulatory care, generic medicines were more susceptible to shortages than branded medicines.[7] Longer shortages were associated with modest increases in medicine prices (6%-14%). In Canada, markets in which only a single generic manufacturer was present were more vulnerable to a medicine shortage.[6] In France, older markets are affected more frequently.[5] Injectable medicines or those with nonsolid routes of administration are more susceptible to shortages[5,7,16] as are those for certain disease areas.

The objective of this study is to describe the extent to which medicine shortages available in the Swiss health system led to reduced access to pharmaceutical care. We combined data from a nationwide medicine shortage repository with additional data sources. We performed cross-sectional and longitudinal analyses to identify the incidence, duration, intensity, and scope of shortages. To assess the economic impact of shortages, we calculated changes in pharmaceutical expenditure. We then assess how supply-chain interruptions paired with spikes in demand for certain products during the first lockdown period of the COVID-19 pandemic affected access.

In studying medicine shortages in the Swiss health system, we can provide evidence about the effects on access to medicine and economic consequences for healthcare systems equally vulnerable to shortages. Although some of the largest pharmaceutical companies are located in Switzerland,[18] most medicines are produced and supplied from outside.[19] Similar to other countries of Europe and North America, Switzerland relies on the production of active ingredients, most of which is supplied by countries of Asia. The Swiss pharmaceutical market is relatively small and requires extensive regulatory efforts. Considering transaction cost to market products, the number of manufacturers of the same active ingredient may be small.

Methods

Study Design and Data

We combined data from 3 sources to perform retrospective cross-sectional and longitudinal analyses on the effect of medicine shortages, taking into account their incidence, duration, intensity, and cost.

We used data from a national repository on medicine shortages (*drugshortage.ch*) between January 2015 and June 23, 2020. Pharmaceutical manufacturers, physicians, and pharmacists voluntarily report shortages arising in Switzerland. The data set is maintained by a private company (Martinelli Consulting GmbH). Having access to the largest wholesaler of the country, it validates shortage reports by providers and manufacturers at the wholesale level. The Federal Office for National Economic Supply (BWL) of Switzerland also reports shortages that manufacturers need to report since 2015. These are limited to essential medicines by the institute's definition.[8]

To assess the set of pharmaceutical products, from the reference database (Spezialitätenliste, https://www.listofpharmaceuticalspecialities.ch/),[20] we extracted all prefabricated prescription medicines actively marketed between January 1, 2015, and January 1, 2020. The list is maintained by the Swiss Federal Office of Public Health. It contains medicines reimbursable within mandatory health insurance. Medicines may be added based on an assessment of safety, expedience, effectiveness, and economic viability.[18]

We obtained national-level data about ambulatory care prescriptions at product level from Helsana, 2013 to 2019, the largest health insurer in Switzerland.[21] To link with data from other sources, we collected product-level auxiliary data from HCI Solutions.[22] We considered products listed in the pharmaceutical reference database in 2018, the retail prices of these medicines, and the shortages. We also considered the national volumes of prescriptions for 2017 and 2018.

The data were processed and analyzed using Stata version 16.0 MP and R version 4.0.1. All used data and programs are documented via OSF.

No ethical approval was required for this study.

Selection of Study Sample and Outcomes

Medicines are defined by a Global Trade Identification Number (GTIN). In line with previous research,[6,7] we define one market to include all GTINs of products with the same active ingredient as defined by Anatomical Therapeutic Chemical (ATC) classification system level 5, dosage form, route of administration, and strength. We identified 4119 markets.

We define incidence of a shortage if it lasted for at least 14 days as defined by the Swiss BWL and other countries. To identify the intensity of medicine shortages across time and markets, we assessed the proportion of options by number of GTINs reported on shortage within the same market. Per market, we classified whether at any point in time a shortage was reported for none of the options in the market, some, or all options. To assess shortages occurring in the context of the COVID-19 pandemic, we identified newly reported shortages after January 23, 2020, the worldwide first occurrence of lockdown measures in Wuhan, China. In the following months, supply chains were heavily disturbed, thereby leading to increased reporting of shortages and stockpiling by pharmacies and patients.[23,24] We fixed the end of our observation period at June 23, 2020, to cover 5 months of the pandemic. We created longitudinal data of the medicine shortages to identify variation in intensity and duration.

Classification of Shortages

For each market, we collected characteristics of the medicines to classify the incidence of shortages. Using the Swiss BWL classification, per active ingredient defined by ATC level 5, we identified whether a medicine was defined as essential. Thus, 161 active ingredients (699 markets) are considered essential, whereas 1033 active ingredients (3430 markets) are not.

To reflect market structure, we assessed several characteristics: average weighted ex-factory price, marketing status, the type of medicine, and co-payment status. Using manufacturer name, we identified how many different companies were selling the medicine within the same market. To assess whether products are available from both brand name and generic manufacturers or only one, we identified the multisource market status. As within selected multisource markets, for the prescriptions covered by the mandatory health insurance, co-payments are increased from 10% to 20% for selected products that are more expensive than their generic counterparts. We also calculated the proportion of products subject to the higher co-payment.[18] Finally, we identified whether the medicine was a small molecule or a biologic medicine.[25]

Pharmaceutical Expenditure Attributable to Medicine Shortages

To identify the economic consequences of shortages, we calculated the changes in pharmaceutical expenditure[26] attributable to shortages in 2018, the year with the highest number of reported shortages. We reflect changes in expenditure identifying price differences between options subject to a shortage and the average prices of options available. To determine the quantity, we calculated the number of prescriptions that were unable to be supplied during the shortage in the affected markets. To calculate the potential number of prescriptions affected by the shortage, we calculated the proportion of prescriptions of that market by accounting for the number of days in 2018 for which a shortage was reported and the proportion of options for which a shortage was reported. We assume that during the period of the shortage, the prescription volume provided is fully replaced. We assume that all prescriptions of medicines subject to shortage are substituted by the best available alternative given full insurance coverage. Because the shortage itself will most likely affect the quantity of a product in a market (similar to the construction of price indexes[27]), we calculated separate estimates on prescription volumes of 2017 and 2018.

To quantify changes in pharmaceutical expenditure, we performed 2 analyses: one on markets with a shortage of some options and one on markets with a shortage of all options. If there was only a shortage of some options, we assume that providers, including physicians and pharmacists, would have first turned to options still available in the same market. These may have included options of the same dosage form and strength but varying numbers of tablets. If all options of the same market were not available, we assume that providers turned to options available for the same active ingredient at ATC level 5 (see Appendix 1 found at https://doi.org/10.1016/j.jval.2021.12.017).

Difference-in-Difference Analyses

To analyze the consequences of the COVID-19 pandemic, we performed difference-in-difference analyses to identify the effect of the first lockdown on the proportion of medicines reported on shortage.[28,29] We classified medicines by essential medicine status and whether a medicine was suitable for treating the symptoms of COVID-19. We considered the first lockdown period as the intervention. We assumed that this new disease led to spikes in demand, thereby making medicines susceptible to shortages and reducing access to the medicines directly affected by the pandemic. We used a list of medicine classes reported by IQVIA in June 2020 to classify medicine treatments used to alleviate symptoms of COVID-19 by ATC 3 groups.[30] As a potential outcome, we consider the proportion of options on shortage. Event studies to assess the parallel trends assumptions indicate that the time trends in the proportion of options on shortage in treatment and control groups were not significantly different before the start of the lockdown period (Appendix 2 found at https://doi.org/10.1016/j.jval.2021.12.017).

Results

Incidence of Shortages

Of the 4119 markets identified, 1964 (47.68%) were never on shortage between 2015 and 2020 (Table 1). In 1336 markets (32.43%), some options were short—but never the full range. In 819 markets (19.88%), all options were on shortage for at least 14 days. Comparing market characteristics, we found several differences regarding shortage status. Essential medicines for which it is mandatory to report a shortage as defined by authorities in Switzerland represent 17.62% of all markets. In 50.37% of markets, a shortage was reported at some point in time; thus, shortages are not limited to essential medicines. When considering the type of the medicine, most shortages are reported for small molecules. Of the 412 biotechnological treatments identified, 204 were on shortage compared with 1179 small molecule markets. Multisource markets that contain several brand name and generic versions of the same medicine were more frequently affected than single source markets that mostly reflect brand name markets. Considering the distribution across different therapeutic groups, shortages were spread across the universe of active ingredients available with few exceptions. Shortages were more frequent in treatments for the metabolic system, the nervous system, the heart and circulation, and infectious diseases (Table 1).

During our observation period, the volume of prescriptions affected by medicine shortages strongly increased (Table 1). The market volume in which at least 1 product of a market was reported short covered 50% of the total prescription medicines market. Hence, a large part of the market is susceptible to shortages. Between 2015 and 2019, in markets in which some options were on shortage, the number of prescriptions increased from 30.51 million to 75.62 million. In markets with all options short, the number increased from 1.03 to 3.57 million prescriptions. The proportion of prescriptions that can be effectively attributed to the options reported on shortage is much smaller but equally increasing. The volume of prescriptions of medicines for which some options were short increased from 1.24 million to 16.07 million. The volume of prescriptions for which all options were short increased from 1.08 million to 1.87 million between 2015 and 2019.

Markets that never experienced a shortage are supplied by a significantly higher number of manufacturers (ie, 0.93 more, on average). The average ex-factory price of the alternatives is lower (457.64 CHF), and 8.07% more products in the same market are subject to higher co-payments by patients (Fig. 1). Classifying active ingredients by whether both generic and brand name versions are available demonstrates that shortages are more frequent in multisource markets. When classifying markets by whether, at any point in time, all options were short or not, we do not find significant differences.

Duration and Intensity of Shortages

Of the 1629 markets with average shortages longer than 14 days and excluding outliers, the median shortage duration was 83.75 days (mean duration 133.18 days; SD 146.78). The periods in which all options within the same market were reported short were much longer (Fig. 2). The median duration of these episodes was 8 months (mean 16.69 months). The mean duration of unavailability of all options of essential medicines was on average 2.30 months longer than for all other medicines although not significantly different (P = .05, chi-squared test).

Pharmaceutical Expenditure Attributed to Medicine Shortages

In 2018, the total net change in pharmaceutical expenditure attributed to medicine shortages amounted to CHF 17.00 million (€15.63 million) considering market volumes of 2018 and retail prices (Table 2). This is the amount to be paid by the mandatory health insurance if prescriptions reported on shortage were substituted by the alternative options. Of these, a reduction in expenditure of CHF 0.19 million was attributed to markets with some options on shortage. CHF 17.19 million were attributed to markets with all options short. These estimates are lower (CHF

Table 1. Incidence of shortages in Switzerland, 2015-2020.

	Medicine shortage status 2015-2020			
	No options on shortage	Some options on shortage	All options on shortage*	Total
Essential medicine status				
Nonessential medicine	1612	1097	711	3420
Essential medicine	352	239	108	699
Type of medicine				
Biotech	208	116	88	412
Small molecule	1150	745	434	2329
Type not classified	606	475	297	1378
Multisource market				
Branded/generics only	1253	577	513	2343
Multisource market	522	738	301	1561
Market not classified	189	21	5	215
Therapeutic group				
Metabolic system	631	347	233	1211
Nervous system	355	285	174	814
Cardiovascular system	128	230	101	459
Infectious diseases	171	137	102	410
Blood	175	36	31	242
Dermatological	63	62	65	190
Gastroenterological	45	64	23	132
Respiratory system	55	27	39	121
Renal system	40	34	12	86
Ophthalmologic	24	39	17	80
Diagnostic	35	17	0	52
Otorhinolaryngologic	17	13	9	39
Metabolic system	5	5	8	18
Gynecology	7	9	0	16
Other	194	30	5	229
Volume in million prescriptions				
2015	73.233	30.513	1.034	104.780
2016	47.767	58.729	2.262	108.758
2017	41.787	65.665	3.065	110.517
2018	39.164	69.281	3.350	111.795
2019	33.515	75.620	3.569	112.703
Volume in million prescriptions of options reported on shortage				
2015	-	1.242	0.121	1.079
2016	-	4.988	1.505	4.514
2017	-	9.215	2.130	8.261
2018	-	13.532	2.141	12.406
2019	-	16.065	1.866	14.028
Markets (N)	1964	1336	819	4119

Note. One market includes all products identified by Global Trade Identification Number (GTIN) with the same active ingredient as defined by Anatomical Therapeutic Chemical classification system level 5, dosage form, route of administration, and strength. Essential medicines as defined by the Swiss Federal Office of National Supply (BWL). Data were obtained from repository of reported shortages (*drugshortage.ch*), the pharmaceutical reference database (*Spezialitätenliste*), and prescription volumes at national level provided by Helsana health insurance.
BWL indicates Bundesamt für wirtschaftliche Landesversorgung.

Figure 1. Characteristics of markets by shortage status. Error bars indicate 95% confidence intervals. Numbers above bars reflect number of markets. Number of markets reported in the panel of ex-factory price per options are lower due to missing price data for selected markets. Data were obtained from repository of reported shortages (drugshortage.ch) and pharmaceutical reference database (*Spezialitätenliste*), 2015 to June 2020.

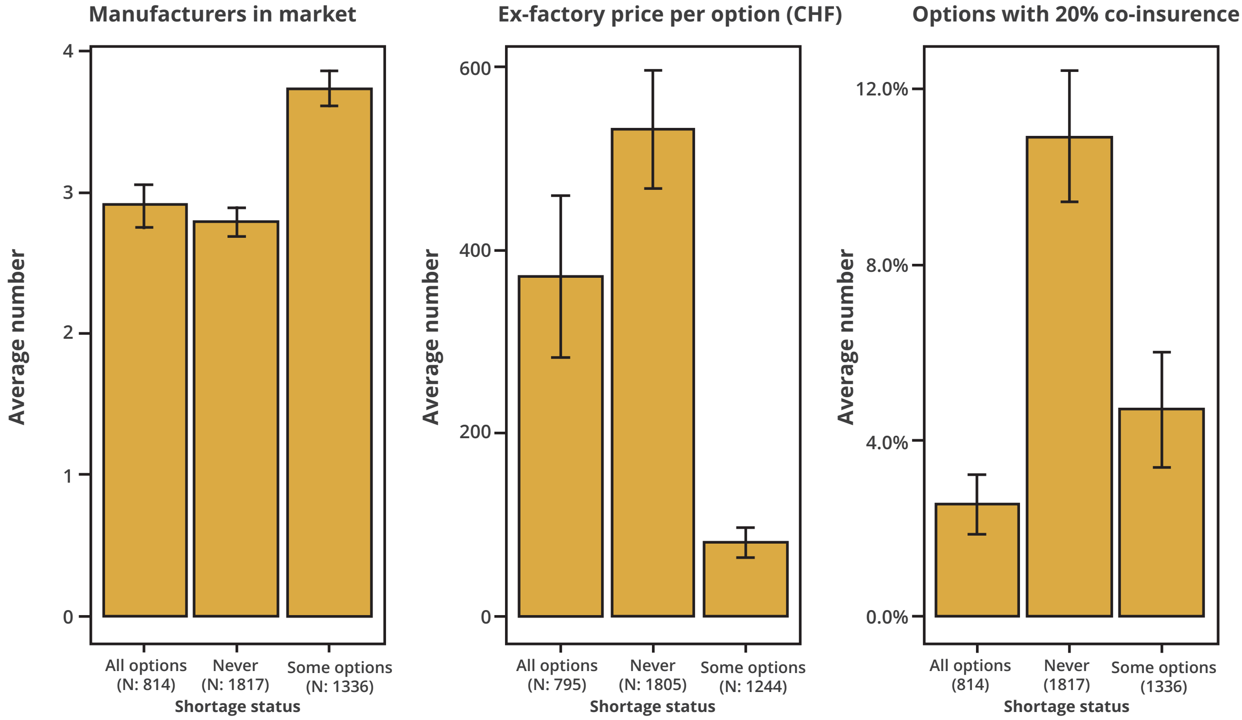
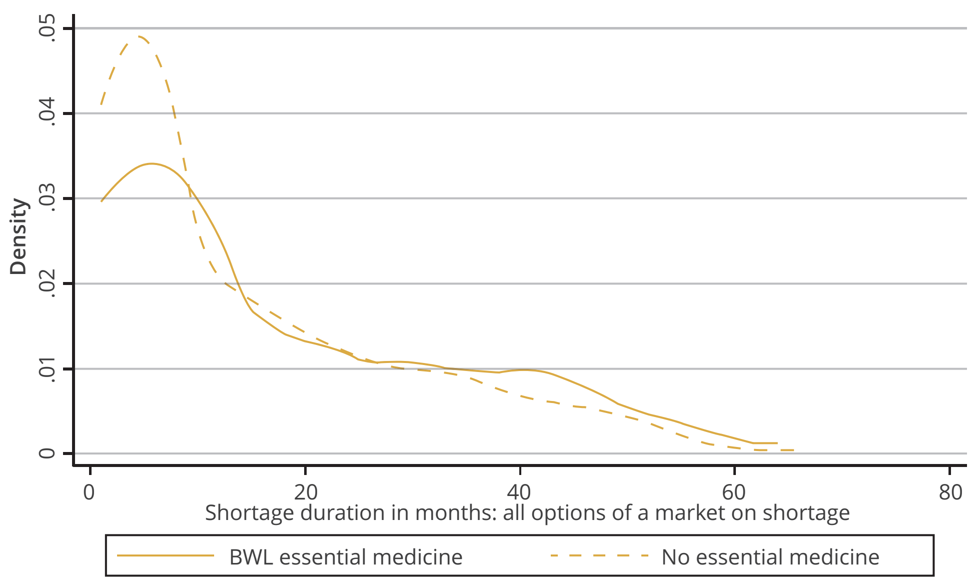


Figure 2. Kernel densities of duration of shortages: markets when all options of the same market are not available, by essential medicine status. Essential medicines as defined by the Swiss Federal Office of National Supply (BWL). Data were obtained from repository of reported shortages (drugshortage.ch) and pharmaceutical reference database (*Spezialitätenliste*), 2015 to June 2020. The figure reports kernel densities of shortage durations of 777 markets in which all options of a market were on shortage during the observation period.



BWL indicates Bundesamt für wirtschaftliche Landesversorgung.

Table 2. Pharmaceutical expenditure attributable to medicine shortages, 2018.

Category	Pharmaceutical expenditure attributed to medicine shortages, 2018				Markets with $\overline{p}_{As}>p_s$	Markets with $\overline{p}_{As}<p_s$
	2017 Market volumes		2018 Market volumes		N	N
	(u)	(w)	(u)	(w)	(%)	(%)
All options short	16.73	14.89	20.35	17.19	359 (57.17)	269 (42.83)
Some options short	−9.81	−1.92	−10.88	−0.19	387 (57.68)	284 (42.32)
Total	6.92	12.97	9.47	17.00	746 (57.43)	553 (42.57)

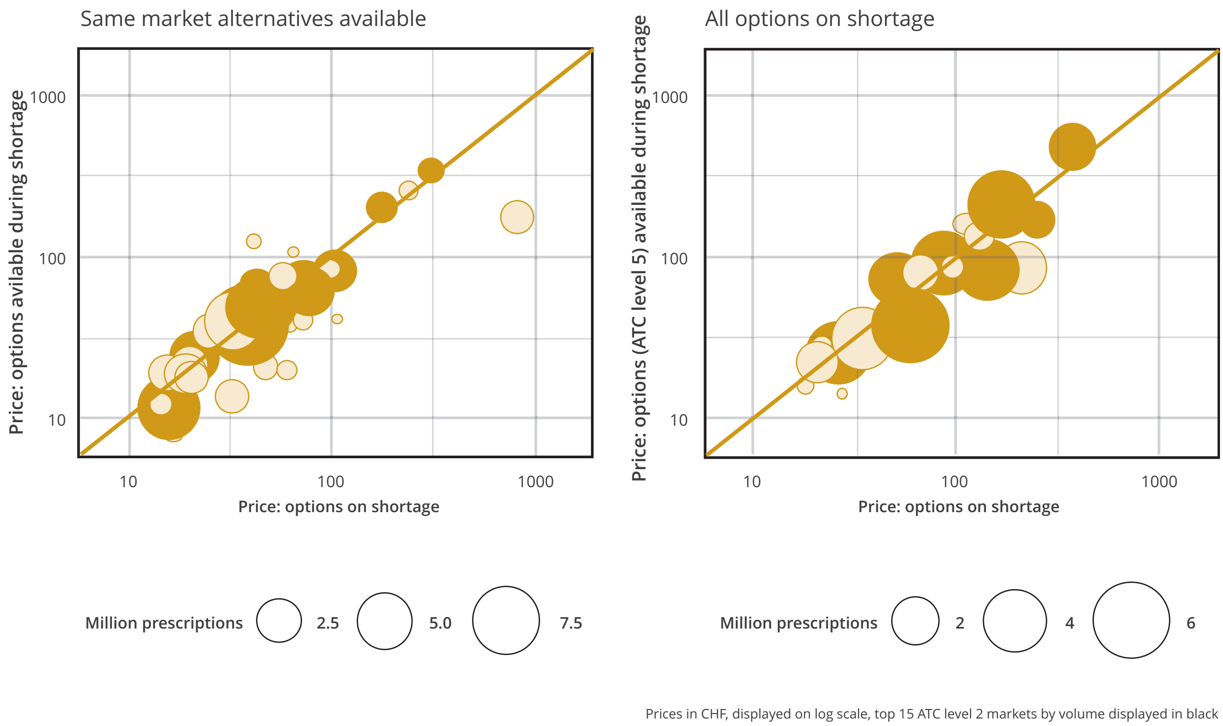
Note. (w): weighted indicates that average prices of available options were weighted by prescription volumes of the available options in 2018. (u): unweighted indicates that average prices of available options were calculated as the arithmetic mean of prices. $\overline{p}_{As}$: weighted average price of the alternative medicine; p_s : price of the medicine on shortage. Data were obtained from repository of reported shortages (*drugshortage.ch*), the pharmaceutical reference database (*Spezialitätenliste*), and prescription volumes provided by Helsana health insurance.

12.97 million) when we consider prescription volumes from 2017 to calculate changes in pharmaceutical expenditure of shortages reported in 2018. Our estimates are lower when we do not weight the prices of the options available during a shortage by their market volumes.

We related these changes in expenditure to lower and higher prices of alternative options (Table 2). In 57.43% of markets in 2018, the price difference between options that remained available in the market compared with options on shortage was negative. In the other 42.57%, the price difference was positive. Comparing whether only some or all options were on shortage, the proportions of markets with higher prices of the alternatives were roughly equal to markets with lower alternative prices.

Figure 3 displays the variability in prices of the alternatives across medicine classes by shortage status. We compared the prices of options on shortage (x-axis) with prices of options still available (y-axis). The 45-degree line reflects situations in which net prices are zero, leading pharmaceutical expenditure not to change because of medicine shortages. In cases above the 45-degree line, the prices of the options still available are higher, thereby suggesting that pharmaceutical expenditure is higher during a shortage. In cases below the 45-degree line, the prices of the options still available are lower, thereby suggesting that pharmaceutical expenditure is lower. Figure 3 highlights that markets of all sizes are equally affected. There is no clear pattern to determine whether markets belong to high-volume markets. The variability in

Figure 3. Average retail prices of medicine options subject to shortage and prices of same market alternatives. One bubble indicates an ATC class level 2. ATC classes that belong to the 15 drug classes are indicated in dark. All other drug classes are indicated in light. Data were obtained from repository of reported shortages (drugshortage.ch), the pharmaceutical reference database (*Spezialitätenliste*), and prescription volumes provided by Helsana health insurance.



ATC indicates Anatomical Therapeutic Chemical.

Table 3. Difference-in-difference analysis of shortages by COVID, including fixed effects, and proportion of options on shortage by market as dependent variable.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Essential medicine	−0.00275	0.0358*	−0.0535†	−0.0759†				
	(0.003)	(0.004)	(0.010)	(0.030)				
Essential medicine X Lockdown	−0.0227*	−0.0240*	−0.0353*	−0.0166				
	(0.005)	(0.005)	(0.011)	(0.025)				
COVID-19 treatment					−0.0211*	−0.0604*	−0.00478	−0.0174
					(0.003)	(0.009)	(0.013)	(0.034)
COVID-19 X Lockdown					0.0132†	0.0125†	0.0334	0.0229
					(0.005)	(0.005)	(0.018)	(0.042)
Lockdown period	−0.0810*	−0.0797*	−0.0756*	−0.0681*	−0.0873*	−0.0862*	−0.0918*	−0.0782*
	(0.005)	(0.005)	(0.012)	(0.018)	(0.005)	(0.005)	(0.013)	(0.011)
Constant	0.106*	0.208*	0.0920‡	0.163†	0.109*	0.209*	0.0957*	0.168†
	(0.005)	(0.034)	(0.029)	(0.073)	(0.005)	(0.034)	(0.029)	(0.073)
Time (mo) FE	Yes	Yes	Yes	No	Yes	Yes	Yes	No
Product class FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Volume weights	No	No	Yes	Yes	No	No	Yes	Yes
2 × 2 DID	No	No	No	Yes	No	No	No	Yes
R2	0.0139	0.0663	0.0682	0.0963	0.0143	0.0662	0.0663	0.0913
RMSE	0.267	0.260	0.278	0.235	0.267	0.260	0.279	0.236
F-statistic	119.9	83.19	52.86	8.211	116.8	81.09	51.30	7.866
Observations	67 601	67 601	66 707	7514	67 601	67 601	66 707	7514

Note. Robust standard errors in parentheses. Lockdown period refers to January 23, 2020, to June 23, 2020. Essential medicines as defined by the Swiss Federal Office of National Supply (BWL). COVID-19 treatments are medicines used to treat symptoms of COVID-19.[31] Product class fixed effects are at ATC level 2. Volume weights are defined by prescription volumes of the medicines at ATC 5 level in 2019. 2 × 2 DID refers to regressions where the pre- and post-treatment periods are collapsed into one observation, reflecting averages of the same time periods (January to June) in 2019 and 2020. Data were obtained from repository of reported shortages (*drugshortage.ch*) and pharmaceutical reference database (*Spezialitätenliste*), 2015 to June 2020.
ATC indicates Anatomical Therapeutic Chemical; BWL, Bundesamt für wirtschaftliche Landesversorgung; DID, difference in difference; FE, fixed effects; RMSE, root mean square deviation.
*$P < .001$.
†$P < .05$.
‡$P < .01$.

the effect on expenditure becomes evident when accounting for market volumes to calculate the expenditure (Appendix Table 1 found at https://doi.org/10.1016/j.jval.2021.12.017).

Shortages During the First Lockdown of the COVID-19 Pandemic

Finally, we turn to the spike in demand and interruptions in the supply chain caused by the COVID-19 pandemic. Estimates show that the average proportion of options on shortage decreased significantly by 6.81% to 8.10% ($P < .001$) during the lockdown period (columns 1 through 4 of Table 3). After the initiation of the lockdown in China, the proportion of packages of essential medicines reported to be short was significantly reduced (1.66%-3.53% on average) compared with nonessential medicines, although this effect was not significant across all model specifications. Considering that on average 10.6% to 20.8% of medicines are reported on shortage, effect sizes are substantial (10.18%-38.37% of the options on shortage).

The second comparison accounts for medicines eligible for treating patients with COVID-19 relative to all other medicines (columns 5-8, Table 3). Difference-in-difference estimates suggest that during the first lockdown period, the proportion of options on shortage that are considered treatments of COVID-19 slightly increased (1.25% to 3.34%) compared with all other medicines. Comparing this effect with the average proportion of options on shortage (9.57%-10.9%), this effect reflects 5.98% to 34.9% on average. Similar to the models that compare essential and nonessential medicines, we find that during the lockdown, the proportion of options reported on shortage decreased significantly (7.82%-8.63% points). This may be due to a decrease in demand. Assuming that nonessential medicines and medicines used as COVID-19 treatments were unaffected by the lockdown measures, the event studies that we performed to assess the parallel trends assumption suggested no significant difference in the periods before the first lockdown measure in the treated and control groups (Appendix Table 1 found at https://doi.org/10.1016/j.jval.2021.12.017).

Discussion

Using national-level data of the medicines available and shortages based on voluntary reports, our retrospective analysis for Switzerland suggests that approximately half of the prescription market was subject to shortages between 2015 and June

2020. Periods in which there is temporarily no access to all medicines of a market are much longer than when access is only partially limited. Medicine shortages have substantially reduced access to pharmaceutical care in some markets. Nevertheless, we did not find differences by essential medicine status. This reduced access to medicines can have numerous consequences for patient health (eg, discontinuation of effective treatments, hospitalization, or even death). During the first lockdown of the COVID-19 pandemic, it appears that the number of essential medicines on shortage decreased. This decrease may have been caused by a decrease in demand induced by the reduction in physician visits and hospital stays during the first lockdown. The increase in shortages of COVID-19 treatments is inconclusive.

The estimates of the pharmaceutical expenditure attributed to shortages need to be discussed in the light of their relative and absolute effect for health insurance and the static efficiency of the pharmaceutical market. The additional pharmaceutical expenditure of CHF 6.92 to 17 million (€6.36-15.63 million) that we identified is relatively small compared with the total of CHF 7.6 billion in 2018 (<0.3%).[21] Assuming no change in quantity, this estimate equals the expected change in pharmaceutical expenditure. The large variability of changes across medicine classes suggests that savings also occur because expenditure decreased in 50% of shortage episodes as result of the lower prices of the alternatives. Furthermore, we assessed only the potential direct effects on expenditure. Substitutive nonpharmaceutical treatments and excess time and resources used to adjust therapies are not considered. Companies might adjust prices if some markets experience a shortage. This is especially relevant in monopolistic markets and may affect static efficiency.

We demonstrate that prices of medicines where shortages never occur are significantly higher than markets reporting shortages of some or all options. This shows that for higher priced drugs, manufacturers also seem to exert effort to ensure the resilience of their supply chain, something that may not be considered in lower-priced markets.[31] In addition, drugshortage.ch has become an important resource for industry to obtain information about ongoing shortages.

Similar to evidence from outside Switzerland, shortages affect most pharmaceutical therapies such that the issue is not confined to national health system–related regulations. Although we do not directly compare reported shortages, the characterizations of incidence, duration, and market characteristics are similar.[5-7,16] Depending on the observation period and definition of a market, studies report that 25% to 50% of markets have been affected by shortages. For shortage duration, the Swiss numbers we report are similar to those in Canada where reporting of shortages is mandatory for the full range of medicines starting from 2017 (155.3 days [SD 146.2] in Canada compared with 133.18 days [SD 146.78] in Switzerland).[6] In Canada, compared with the United States, the duration of shortages decreased when the mandate was present.[32] For France, a similar increase in reported shortages is documented for 2017 to 2018.[5]

This study is not without limitations. Our analysis is based on shortages reported and validated using wholesale data. We were unable to effectively assess the number of patients who did not receive a prescription caused by a shortage. This may partly explain the strong increase in shortages between 2015 and 2017 as healthcare providers and the industry become aware of the problem. Another effect of using reported shortages is that we cannot account for the availability of products reported under shortage in stock locally. Nevertheless, we can assume that in the case of shortages of 14 days or longer, many pharmacies will be low or out of supply. In addition, the 14 days cutoff may have different implications for patients depending on a medicine's dosing regimen. We were also unable to assess decisions to switch to options outside the same market or to discontinue therapy. We cannot account for the cost of hospital episodes arising from discontinued medicine treatments. Our analysis disregards the potential indirect cost caused to patients and additional transaction costs for pharmacists and physicians who have to cope with shortages.[33] Finally, our classification of manufacturers does not allow us to identify whether the same active ingredient of different manufacturers is produced at the same manufacturing site, which is frequently the case. We also cannot rule out that the reporting quality in the medicine shortage database was different during the first COVID-19 pandemic compared with the periods before.

Conclusion

Shortages have significantly reduced access to medicines by the evenness of their availability across time and ability to pay, especially when no alternative is available in the same market. Our study confirms previous evidence from outside Switzerland that has reported substantial increases in medicine shortages. Although the direct consequences on pharmaceutical expenditure we identify based on our approach are small, the extent to which some pharmaceutical markets are at risk of shortages that leave no alternative option available is considerable. Policy makers need to bear in mind that shortages are not confined to essential medicines and that national policies that target certain therapeutic areas may not be successful.

Supplemental Materials

Supplementary data associated with this article can be found in the online version at https://doi.org/10.1016/j.jval.2021.12.017.

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REFERENCES

1. Vyas M, de Vries EGE, Casali PG, Tabernero J. Shortages of inexpensive essential medicines. *Lancet Oncol*. 2019;20(5):e224–e225.
2. Dill S, Ahn J. Drug shortages in developed countries—reasons, therapeutic consequences, and handling. *Eur J Clin Pharmacol*. 2014;70(12):1405–1412.
3. Woodcock J, Wosinska M. Economic and technological drivers of generic sterile injectable drug shortages. *Clin Pharmacol Ther*. 2013;93(2):170–176.
4. De Weerdt E, Simoens S, Hombroeckx L, Casteels M, Huys I. Causes of drug shortages in the legal pharmaceutical framework. *Regul Toxicol Pharmacol*. 2015;71(2):251–258.
5. Benhabib A, Ioughlissen S, Ratignier-Carbonneil C, Maison P. The French reporting system for drug shortages: description and trends from 2012 to 2018: an observational retrospective study. *BMJ Open*. 2020;10(3):e034033.
6. Zhang W, Guh DP, Sun H, et al. Factors associated with drug shortages in Canada: a retrospective cohort study. *CMAJ Open*. 2020;8(3):E535–E544.
7. Dave CV, Pawar A, Fox ER, Brill G, Kesselheim AS. Predictors of drug shortages and association with generic drug prices: a retrospective cohort study. *Value Health*. 2018;21(11):1286–1290.
8. Bundesamt für wirtschaftliche Landesversorgung BWL (Federal Office for National Economic Supply), 2020. Meldestelle für lebenswichtige Humanarzneimittel Bericht 2019 (Notification Center for Essential Human Medicines Report 2019). Bern, Switzerland. https://www.bwl.admin.ch/dam/bwl/de/dokumente/themen/heilmittel/meldestelle/bericht_heilmittelplattform_2019.pdf.download.pdf/Bericht%20HMP%202019.pdf. Accessed February 21, 2022.
9. Alexander GC, Qato DM. Ensuring access to medications in the US during the COVID-19 pandemic. *JAMA*. 2020;324(1):31–32.
10. European Parliament, Directorate-General for Internal Policies of the Union, Laky, Z., Medicine shortage in the EU during the novel coronavirus outbreak, European Parliament, 2020, https://data.europa.eu/doi/10.2861/788214. Accessed February 21, 2022.
11. Jackevicius CA, Krumholz HM, Chong A, et al. Population impact of generic valsartan recall. *Circulation*. 2020;141(5):411–413.
12. Stomberg C. Drug shortages, pricing, and regulatory activity. National Bureau of Economic Research. NBER Working Paper 22912. https://www.nber.org/system/files/working_papers/w22912/w22912.pdf. Accessed February 21, 2022.
13. Costa-Font J, McGuire A, Varol N. Price regulation and relative delays in generic drug adoption. *J Health Econ*. 2014;38:1–9.
14. Cohen J, Faden L, Predaris S, Young B. Patient access to pharmaceuticals: an international comparison. *Eur J Health Econ*. 2007;8(3):253–266.
15. Aday LA, Andersen R. A framework for the study of access to medical care. *Health Serv Res*. 1974;9(3):208–220.
16. Chen SI, Fox ER, Hall MK, et al. Despite federal legislation, shortages of drugs used in acute care settings remain persistent and prolonged. *Health Aff (Millwood)*. 2016;35(5):798–804.
17. Vogler S, Fischer S. How to address medicines shortages: findings from a cross-sectional study of 24 countries. *Health Policy*. 2020;124(12):1287–1296.
18. De Pietro C, Camenzind P, Sturny I, et al. Switzerland - health system review. *Health Syst Transit*. 2015;17(4):1-288, xix.
19. Flury M, Geiger A, Sandmeier H, Wüthrich J. *Pharma-Markt Schweiz 2019 (Swiss pharmaceutical market 2019)*. Interpharma; 2019:120. https://www.interpharma.ch/wp-content/uploads/2020/02/ly_iph.01.19.002_-_pharmamarkt_schweiz_2019_d_web-komprimiert.pdf. Accessed February 21, 2022.
20. Federal Office of Public Health. List of Pharmaceutical Specialities (*Spezialitätenliste (SL) und Geburtsgebrechen-Spezialitätenliste (GGSL)*). http://www.listofpharmaceuticalspecialities.ch/. Accessed December 4, 2020.
21. Schneider R, Schur N, Reinau D, Schwenkglenks M, Christoph M. Helsana-Arzneimittelreport für die Schhweiz 2019. https://www.helsana.ch/de/helsana-gruppe/ueber-uns/gesundheitswissenschaften/arzneimittelreport.html. Accessed July 29, 2020.
22. HCI Solutions. Schema overview article. https://index.hcisolutions.ch/DataDoc/element/ARTICLE. Accessed March 12, 2021.
23. Holden E. US prescription drug supply chains face coronavirus stress test. *The Guardian*. http://www.theguardian.com/world/2020/mar/17/us-prescription-drug-supply-chains-coronavirus. Accessed April 15, 2021.
24. Thomas K. First drug shortage caused by coronavirus, F.D.A. Says. But it won't disclose what drug or where it's made. *The New York Times*. https://www.nytimes.com/2020/02/28/health/drug-coronavirus-shortage.html. Accessed April 28, 2020.
25. Wishart DS, Knox C, Guo AC, et al. DrugBank: a comprehensive resource for in silico drug discovery and exploration. *Nucleic Acids Res*. 2006;34(Database issue):D668–D672.
26. OECD Data. Pharmaceutical spending. http://data.oecd.org/healthres/pharmaceutical-spending.htm. Accessed October 12, 2021.
27. Berndt ER, Cutler DM, Frank RG, Griliches Z, Newhouse JP. Price indexes for medical care goods and services: an overview of measurement issues. National Bureau of Economic Research. http://www.nber.org/papers/w6817. Accessed May 9, 2014.
28. Cunningham S. *Causal Inference: the Mixtape*. New Haven, Connecticut: Yale University Press; 2021.
29. Imbens GW, Rubin DB. *Causal Inference for Statistics, Social, and Biomedical Sciences: an Introduction*. Cambridge: Cambridge University Press; 2015.
30. IQVIA. Pharmamarkt-Entwicklung im Kontext von COVID-19 - Ausgewählte Marktanalysen vor dem Hintergrund der Covid-19-Pandemie - Kalenderwoche 23. Presented at: 2020. https://www.iqvia.com/de-de/locations/germany/library/publications/pharmamarkt-entwicklung-im-kontext-von-covid-19. Accessed March 15, 2022.
31. Duminy L, Grosser SN. Resilience as basis for sustainability: shortages in production supply chains for essential consumer goods. In: Qudrat-Ullah H, ed. *Innovative Solutions for Sustainable Supply Chains. Understanding Complex Systems*. Cham: Springer International Publishing; 2018:223–262.
32. Lee J, Lee HS, Shin H, Krishnan V. Alleviating drug shortages: the role of mandated reporting induced operational transparency. *Manag Sci*. 2021;67(4):2326–2339.
33. Pauwels K, Simoens S, Casteels M, Huys I. Insights into European drug shortages: a survey of hospital pharmacists. *PLoS One*. 2015;10(3):e0119322.